

Repurposing Diffusion-Based Image Generators for Monocular Depth Estimation

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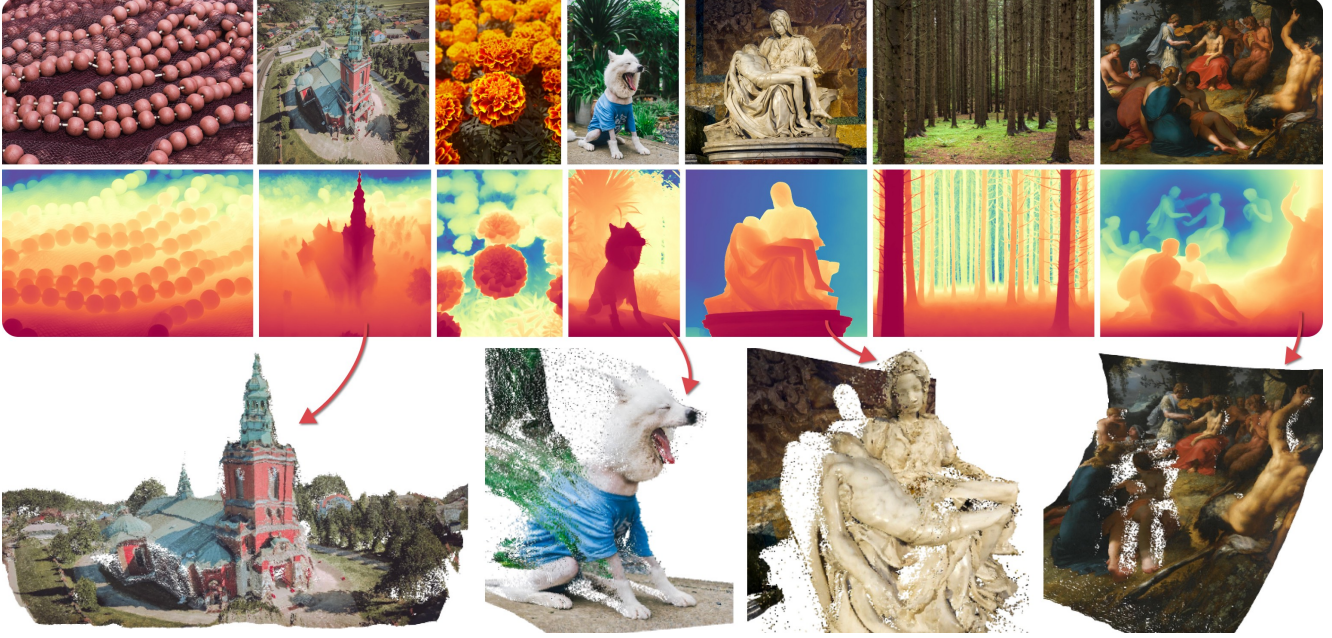


Figure 1. **We present Marigold, a diffusion model and associated fine-tuning protocol for monocular depth estimation.** Its core principle is to leverage the rich visual knowledge stored in modern generative image models. Our model, derived from Stable Diffusion and fine-tuned with synthetic data, can zero-shot transfer to unseen datasets, offering state-of-the-art monocular depth estimation results.

Abstract

Monocular depth estimation is a fundamental computer vision task. Recovering 3D depth from a single image is geometrically ill-posed and requires scene understanding, so it is not surprising that the rise of deep learning has led to a breakthrough. The impressive progress of monocular depth estimators has mirrored the growth in model capacity, from relatively modest CNNs to large Transformer architectures. Still, monocular depth estimators tend to struggle when presented with images with unfamiliar content and layout, since their knowledge of the visual world is restricted by the data seen during training, and challenged by zero-shot generalization to new domains. This motivates us to explore whether the extensive priors captured in recent generative diffusion models can enable better, more generalizable depth estimation. We introduce Marigold, a method for affine-invariant

monocular depth estimation that is derived from Stable Diffusion and retains its rich prior knowledge. The estimator can be fine-tuned in a couple of days on a single GPU using only synthetic training data. It delivers state-of-the-art performance across a wide range of datasets, including over 20% performance gains in specific cases. Project page: <https://marigoldmonodepth.github.io>.

1. Introduction

Monocular depth estimation aims to transform a photographic image into a depth map, *i.e.*, regress a range value for every pixel. The task arises whenever the 3D scene structure is needed, and no direct range or stereo measurements are available. Clearly, undoing the projection from the 3D world to a 2D image is a geometrically ill-posed problem and can